

PDS_PBL_PROJECT

Topic: Restaurant Billing System

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# =====
# RESTAURANT BILLING SYSTEM - DATA ANALYSIS
# =====

# Load dataset
df = pd.read_csv("tips.csv")

print("=====")
print("  RESTAURANT BILLING SYSTEM")
print("    DATA ANALYSIS")
print("=====")

# -----
# BASIC DATA ANALYSIS
# -----

print("\n1. DATASET INFORMATION")
print("-----")
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("Columns:", list(df.columns))
```

```
print( "\nMissing Values:")
print( df.isnull ().sum())
```

```
print( "\nStatistical Summary:")
print( df.describe ())
```

```
# =====
# MODULE 1: BILL ANALYSIS + BAR GRAPH
# =====
```

```
print( "\n\nMODULE 1: BILL ANALYSIS")
print( "-----")
```

```
print( "Total Bill Sum:", round( df[" total_bill "].sum(), 2))
print( "Average Bill:", round( df[" total_bill "].mean (), 2))
print( "Maximum Bill:", round( df[" total_bill "].max(), 2))
print( "Minimum Bill:", round( df[" total_bill "].min(), 2))
```

```
print( "\nAverage Bill by Day:")
print( df.groupby ("day ")[ " total_bill "].mean ().round (2))
```

```
plt.figure (figsize =(7,5))
df.groupby ("day ")[ " total_bill "].mean ().plot (kind="bar")
plt.title ("Average Bill by Day")
plt.xlabel ("Day")
plt.ylabel ("Average Bill")
plt.tight _ layout ()
plt.show ()
```

```

# =====

# MODULE 2: CUSTOMER ANALYSIS + BAR GRAPH

# =====

print( "\n\nMODULE 2: CUSTOMER ANALYSIS")
print( "-----")

print( "Total Customers:", len(df))

print( "\nCustomers by Day:")
print( df["day"].value_counts())

print( "\nAverage Group Size:",
        round(df["size"].mean(), 2))

print( "\nMaximum Group Size:",
        df["size"].max())

plt.figure(figsize=(7,5))
df["day"].value_counts().plot(kind="bar")
plt.title("Number of Customers by Day")
plt.xlabel("Day")
plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()

```

```
# =====  
# MODULE 3: TIP ANALYSIS + BAR GRAPH  
# =====
```

```
print( "\n\nMODULE 3: TIP ANALYSIS")  
print( "-----")
```

```
print( "Total Tip:", round( df["tip" ].sum(), 2))  
print( "Average Tip:", round( df["tip" ].mean(), 2))  
print( "Maximum Tip:", round( df["tip"]. max(), 2))  
print( "Minimum Tip:", round( df["tip" ].min(), 2))
```

```
print( "\nAverage Tip by Day:")  
print( df.groupby ( "day ")[ "tip" ].mean ().round (2))
```

```
plt.figure (figsize =(7,5))  
df.groupby ( "day ")[ "tip" ].mean ().plot (kind="bar")  
plt.title ( "Average Tip by Day")  
plt.xlabel ( "Day")  
plt.ylabel ( "Average Tip")  
plt.tight _ layout ()  
plt.show ()
```

```
# =====  
# MODULE 4: TIME ANALYSIS + PIE GRAPH  
# =====
```

```
print( "\n\nMODULE 4: TIME ANALYSIS")
```

```
print( "-----")
```

```
print( "Lunch Customers:",  
      (df["time"] == "Lunch" ).sum())
```

```
print( "Dinner Customers:",  
      (df["time"] == "Dinner" ).sum())
```

```
print( "\nAverage Bill by Time:")  
print( df.groupby ( "time ")[ "total_bill" ].mean ().round (2))
```

```
plt.figure (figsize =(7,5))  
df["time"]. value_ counts ().plot (  
    kind="pie",  
    autopct ="%1.1f%%"  
)  
plt.title ("Lunch vs Dinner")  
plt.ylabel ("")  
plt.show ()
```

```
# =====  
# MODULE 5: GENDER ANALYSIS + PIE GRAPH  
# =====
```

```
print( "\n\nMODULE 5: GENDER ANALYSIS")  
print( "-----")
```

```
print( "Customers by Gender:")
```

```

print( df["sex"]. value_ counts ())

print( "\nAverage Bill by Gender:")
print( df.groupby ("sex ")[ "total_bill "].mean ().round (2))

plt.figure (figsize =(7,5))
df["sex"]. value_ counts ().plot (
    kind="pie",
    autopct ="%1.1f%%"
)
plt.title ("Customers by Gender")
plt.ylabel ("")
plt.show ()

# =====
# MODULE 6: BILL-TIP RELATIONSHIP
# + SCATTER GRAPH
# =====

print( "\n\nMODULE 6: BILL AND TIP RELATIONSHIP")
print( "-----")

correlation = df["total_bill "].corr (df["tip"])

print( "Correlation between Total Bill and Tip:",
    round( correlation, 3))

print( "\nHighest Bill:")

```

```
print( df.loc [df[" total_bill "].idxmax ()])
```

```
print( "\nHighest Tip:")
```

```
print( df.loc [df["tip" ].idxmax ()])
```

```
plt.figure (figsize =(7,5))
```

```
plt.scatter (df [" total_bill "], df ["tip"])
```

```
plt.title ("Total Bill vs Tip")
```

```
plt.xlabel ("Total Bill")
```

```
plt.ylabel ("Tip")
```

```
plt.tight _layout ()
```

```
plt.show ()
```

```
# =====
```

```
# FINAL ANALYSIS
```

```
# =====
```

```
print("\n\n=====")
```

```
print( "      FINAL ANALYSIS")
```

```
print( "=====")
```

```
print( "Total Revenue/Bill Amount:",  
      round( df[" total_bill "].sum(), 2))
```

```
print( "Average Bill:",  
      round( df[" total_bill "].mean (), 2))
```

```
print( "Total Tip:",
```

```

round( df["tip" ].sum(), 2))

print( "Average Tip:",
      round( df["tip" ].mean (), 2))

print( "Most Frequent Day:",
      df["day" ].mode()[ 0])

print( "Most Common Time:",
      df["time" ].mode()[ 0])

print( "Most Common Group Size:",
      df["size" ].mode()[ 0])

print( "\nData Analysis Completed Successfully!")

```

Output:

```

=====
RESTAURANT BILLING SYSTEM
DATA ANALYSIS
=====

```

1. DATASET INFORMATION

```
-----
```

Number of Rows: 244

Number of Columns: 7

Columns: ['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size']

Missing Values:

```
total_bill  0
tip         0
sex         0
smoker      0
day         0
time        0
size        0
dtype: int64
```

Statistical Summary:

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

MODULE 1: BILL ANALYSIS

Total Bill Sum: 4827.77

Average Bill: 19.79

Maximum Bill: 50.81

Minimum Bill: 3.07

Average Bill by Day:

day

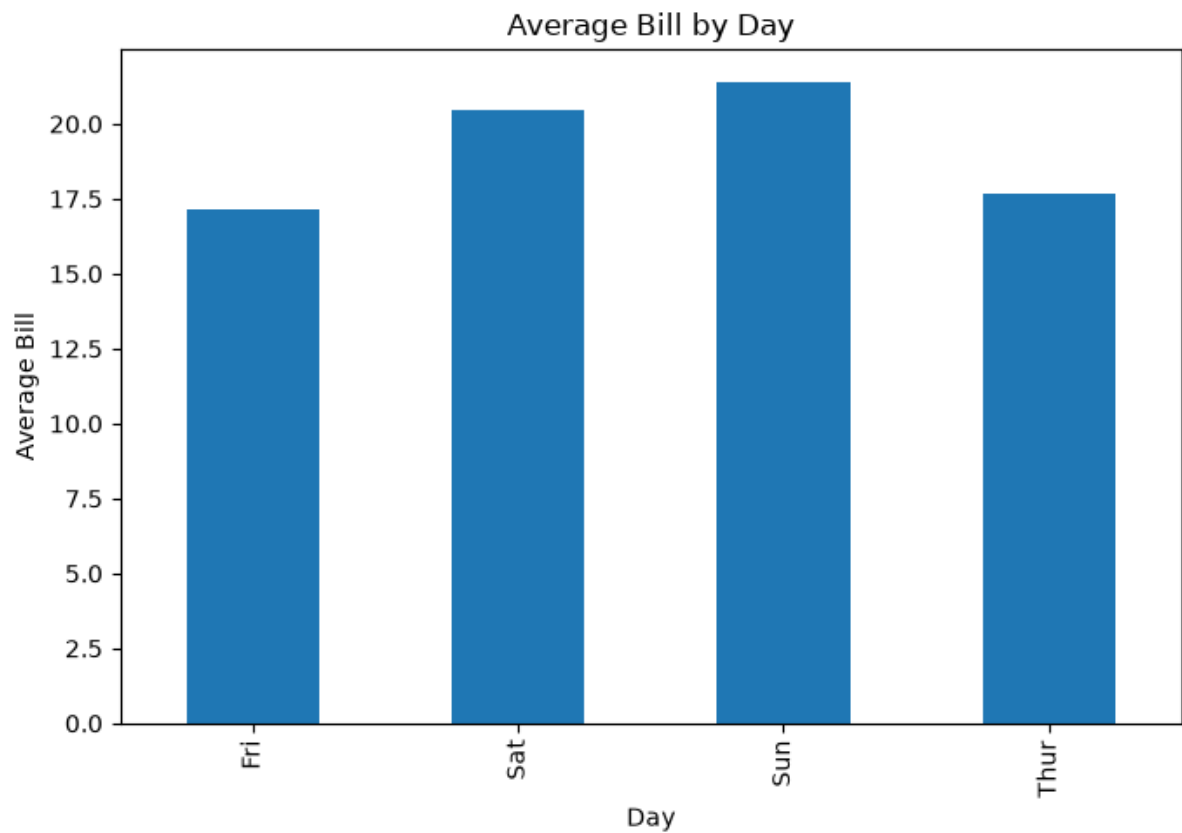
Fri 17.15

Sat 20.44

Sun 21.41

Thur 17.68

Name: total_bill, dtype: float64



MODULE 2: CUSTOMER ANALYSIS

Total Customers: 244

Customers by Day:

day

Sat 87

Sun 76

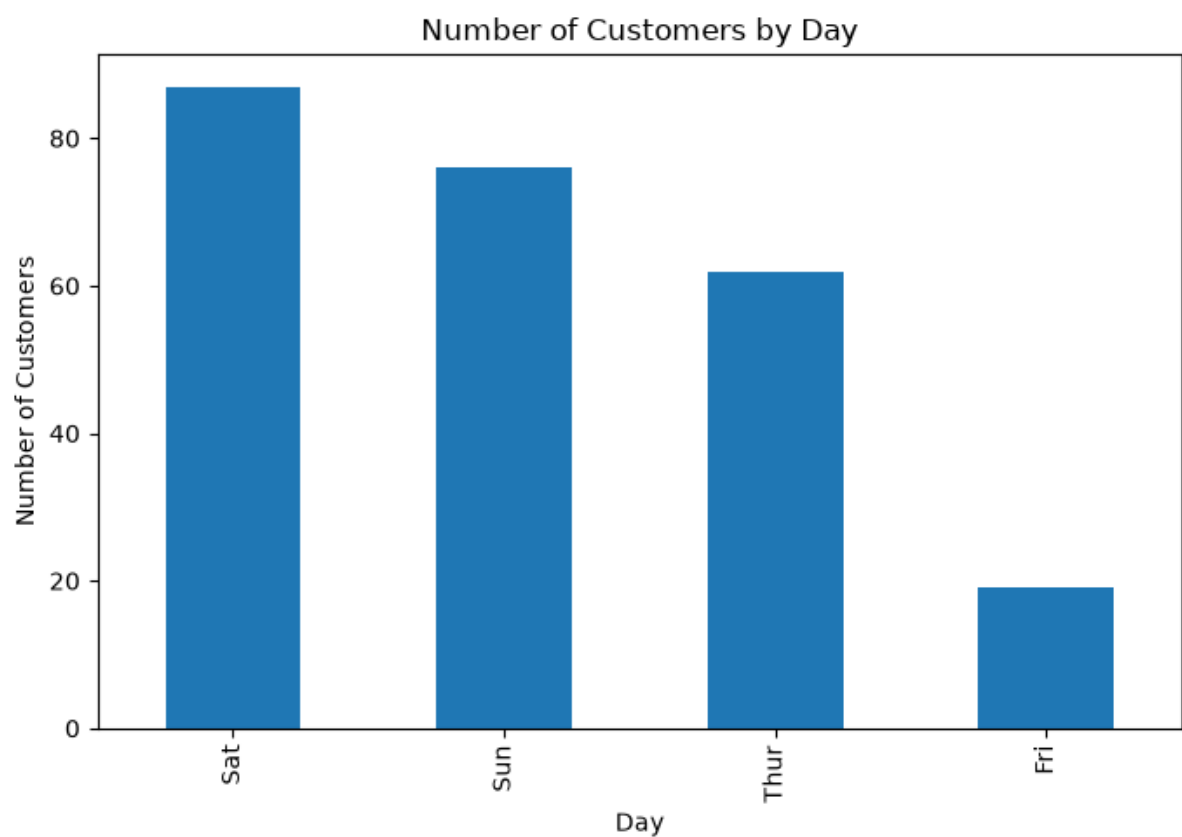
Thur 62

Fri 19

Name: count, dtype: int64

Average Group Size: 2.57

Maximum Group Size: 6



MODULE 3: TIP ANALYSIS

Total Tip: 731.58

Average Tip: 3.0

Maximum Tip: 10.0

Minimum Tip: 1.0

Average Tip by Day:

day

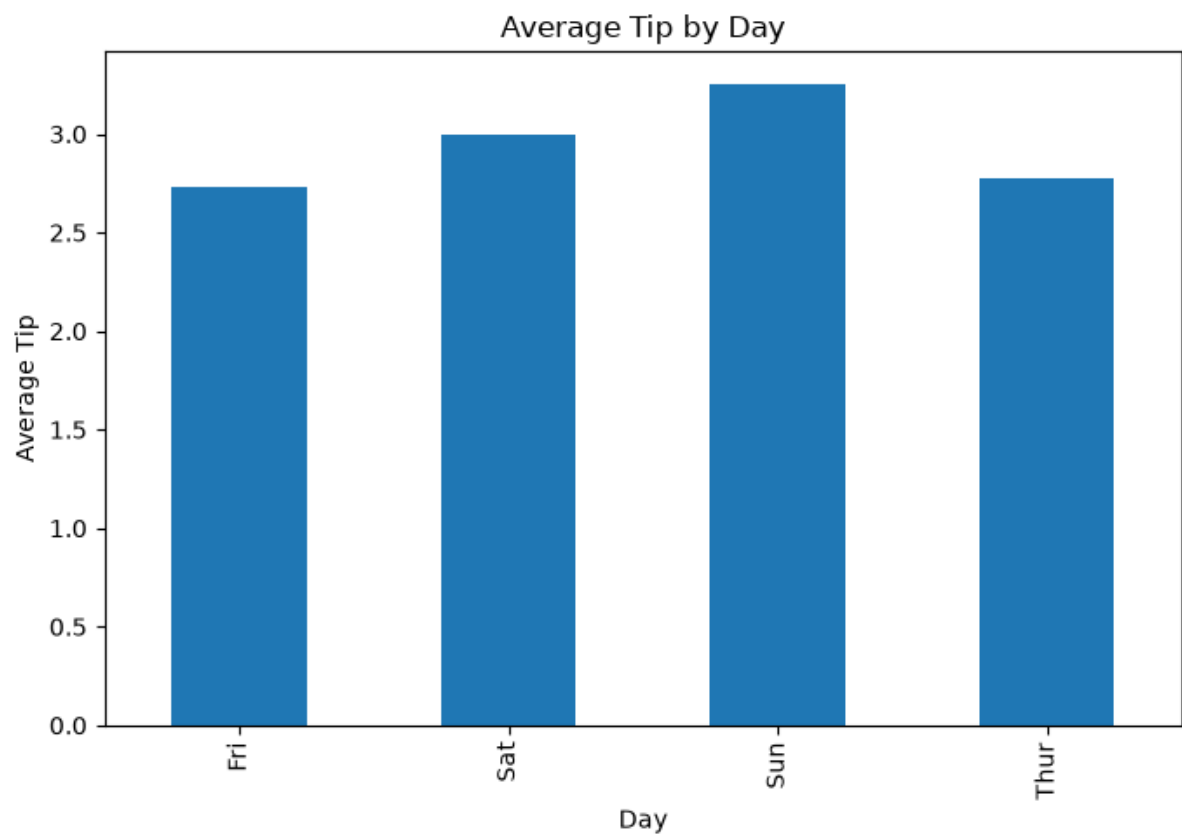
Fri 2.73

Sat 2.99

Sun 3.26

Thur 2.77

Name: tip, dtype: float64



MODULE 4: TIME ANALYSIS

Lunch Customers: 68

Dinner Customers: 176

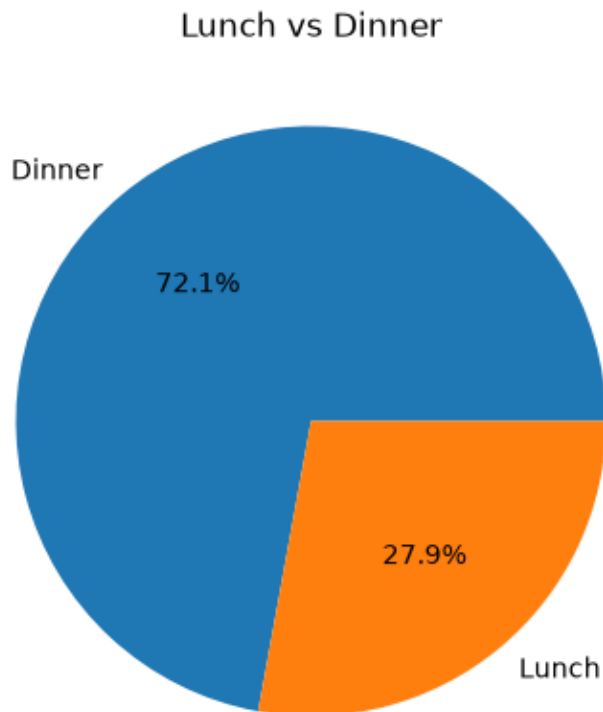
Average Bill by Time:

time

Dinner 20.80

Lunch 17.17

Name: total_bill, dtype: float64



MODULE 5: GENDER ANALYSIS

Customers by Gender:

sex

Male 157

Female 87

Name: count, dtype: int64

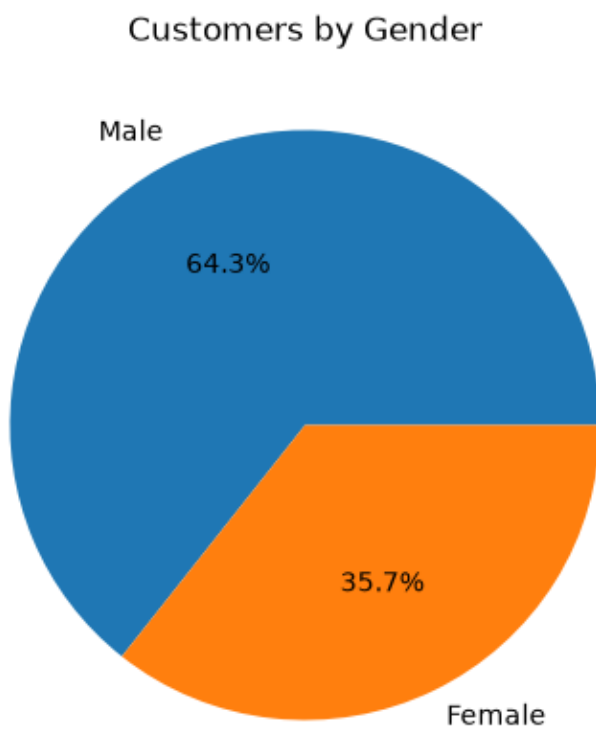
Average Bill by Gender:

sex

Female 18.06

Male 20.74

Name: total_bill, dtype: float64



MODULE 6: BILL AND TIP RELATIONSHIP

Correlation between Total Bill and Tip: 0.676

Highest Bill:

total_bill 50.81

tip 10.0

sex Male

smoker Yes

day Sat
time Dinner
size 3
Name: 170, dtype: object

Highest Tip:

total_bill 50.81

tip 10.0

sex Male

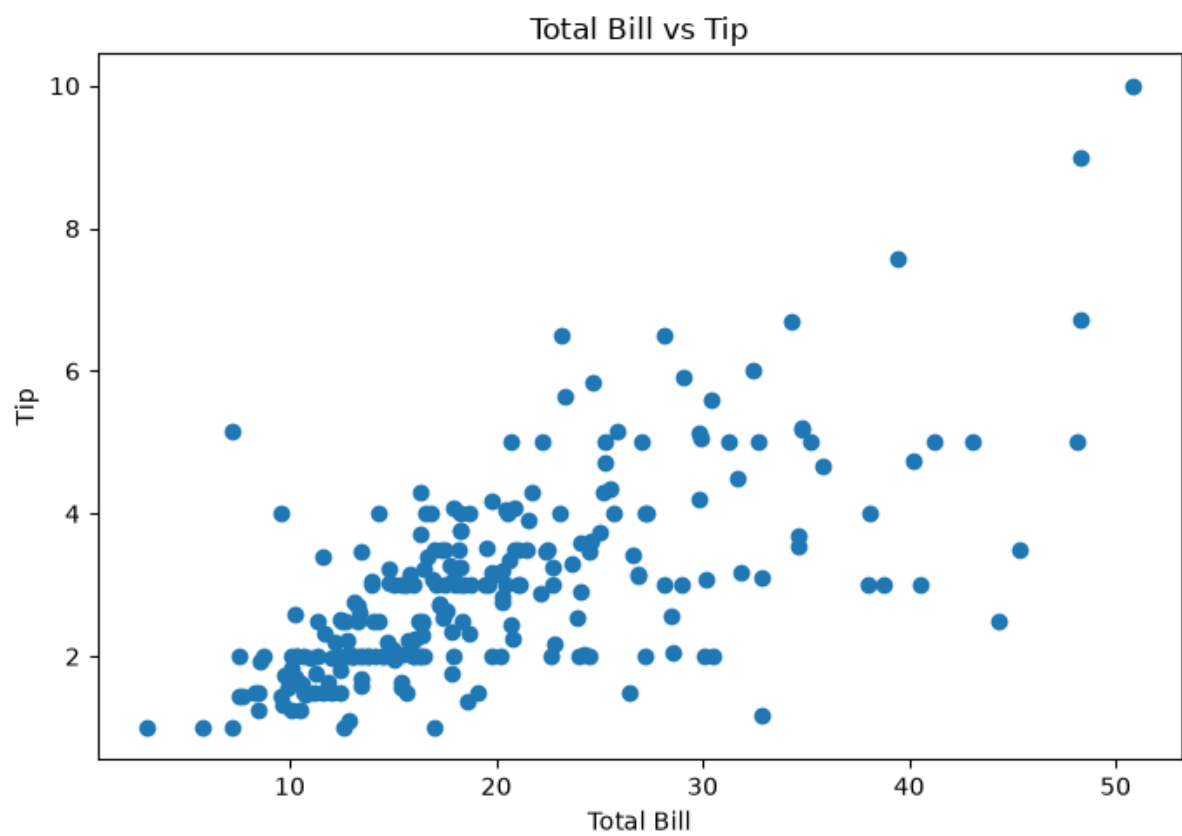
smoker Yes

day Sat

time Dinner

size 3

Name: 170, dtype: object



=====

FINAL ANALYSIS

=====

Total Revenue/Bill Amount: 4827.77

Average Bill: 19.79

Total Tip: 731.58

Average Tip: 3.0

Most Frequent Day: Sat

Most Common Time: Dinner

Most Common Group Size: 2

Data Analysis Completed Successfully!